User Manual: Interactive Fatigue Data Visualizer

Generated from `grphsallpairs.py`

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Contents

1 Introduction

This program is an interactive data visualization tool designed for fatigue analysis. It reads a CSV file containing three columns: `R` (stress ratio), `sigmaM` (mean stress), and `N` (number of cycles). The tool computes derived variables, normalizes them, and displays six scatter plots with an adjustable opacity filter. It is intended to be run in a **Google Colab** environment using `ipywidgets`.

2 Requirements

- Python 3.7 or higher
- Libraries: `pandas`, `numpy`, `matplotlib`, `ipywidgets`, `IPython`
- A Google Colab notebook or a local Jupyter notebook with widget support

3 First Launch

When you run the script, the following actions occur automatically:

1. A default embedded dataset (about 300 rows) is loaded.
2. Six scatter plots are displayed (see Section 6).
3. Three sliders and three buttons appear below the plots.

4 Loading Data

4.1 Default Dataset

The program starts with an internal dataset. No action is needed to load it.

4.2 Loading Your Own CSV File

To analyse your own data:

1. Click the **Seleccionar CSV** button (file upload widget).
2. Select a CSV file from your computer.
3. The file must contain exactly three columns named: `R`, `sigmaM`, `N` (case-sensitive).
4. After selection, the plots update automatically, and all sliders reset to their default values (Center $\alpha = 0.5$, Threshold = 0.2, Min $\alpha = 0.1$).

If the file format is incorrect, an error message will be printed.

4.3 Resetting the Interface

Click the **Leer otro caso** button to reset the sliders to their default positions. A message reminds you to use the file selector to load a new CSV.

5 Controls

5.1 Sliders

Three sliders control the opacity (transparency) of points in all six plots.

`Center` – The central value of the normalized opacity range. Points whose normalized opacity (α) is close to this value become more visible.

`Threshold` – Defines the width of the window around `Center` . Points with α in [Center−Threshold, Center+Threshold] are highlighted (their opacity is increased).

`Min` – The minimum opacity for points outside the selected window. Points inside the window receive a higher opacity, linearly scaled according to their α value.

Normalized opacity α is computed from `sigmaM`:

$$\alpha = \frac{\sigma_M - \min(\sigma_M)}{\max(\sigma_M) - \min(\sigma_M)}$$

5.2 Buttons

`Save PNG` – Saves the current set of six plots as a PNG image. The filename includes a timestamp.

`Save PDF` – Saves the current plots as a PDF vector graphic.

`Leer otro caso` – Resets the sliders to default values and prompts you to upload a new CSV file.

6 Description of the Plots

The figure consists of 2 rows and 3 columns, showing the following relationships:

1. **Top left**: N (cycles) vs. σ_M (mean stress) – log scale on N.
2. **Top middle**: N vs. σ_m (computed as $R \cdot \sigma_M$) – log scale on N.
3. **Top right**: σ_M vs. σ_m – linear scales.
4. **Bottom left**: N vs. X where $X = 1 - R$ – log scale on N.
5. **Bottom middle**: N vs. Y where Y is a normalized version of σ_m:

$$Y = \frac{\sigma_m - \min(\sigma_m)}{\max(\sigma_m) - \min(\sigma_m)}$$

6. **Bottom right**: X vs. Y – linear scales.

All plots have grid lines. The colour of the points is fixed per subplot (blue, green, red, purple, orange, brown). Opacity changes according to the sliders.

7 Example Workflow

1. Run the script. The default dataset appears.
2. Move the `Center` slider to 0.7 and `Threshold` to 0.1. Points with high σ_M become emphasised.
3. Click `Save PNG` to export the current view.
4. Click `Leer otro caso` and then `Seleccionar CSV` to upload your own fatigue data.
5. After loading, the plots refresh with your data and sliders return to 0.5 / 0.2 / 0.1.
6. Adjust the sliders to explore regions of interest.

8 Troubleshooting

- **No plots appear?** Make sure you are running the code in a Jupyter/Colab environment with widget support. Run the cell again.
- **File upload error:** Ensure your CSV has exactly the three columns `R`, `sigmaM`, `N` (no extra spaces, commas as separators).
- **Error: "The 'value' trait is read-only"** – The code already handles this; if you see it, update the `ipywidgets` library or restart the kernel.
- **KeyError: 'name'** – This was fixed in the provided script. If you encounter it, make sure you are using the corrected version of `on_load_case`.
- **Plots are too slow?** Reduce the number of points in your CSV (or use the default dataset which is moderate).

9 How to Obtain the PDF of This Manual

You are reading the LaTeX source. To generate the PDF:

1. Copy the entire LaTeX code into a file named `manual.tex`.
2. Run `pdflatex manual.tex` twice (to resolve references).
3. The output `manual.pdf` will be created.

Alternatively, paste the code into Overleaf and click "Recompile".

10 License and Notes

This tool is free to use and modify. The default dataset is embedded; you may replace it with your own data. For any issues, inspect the console output for error messages.